# Solved Repeated Questions of BT-201

# **Ecology, Biodiversity and Evolution II**

*Made By: Kashif Ashfaq*

*M.Sc Zoology 4th Semester*

1. Succession in the intertidal zone requires 1 to 3 years.
2. Highest rates of terrestrial primary production occur under warm, moist conditions
3. The refuges that promote the persistence of hosts and prey include secure places to which the exploiter has limited access.
5. The environment limits population growth by changing birth and death rates.
6. An environment in which resources are patchy also fosters clumped distributions.
7. This more complex level of social behavior, which is considered to be the pinnacle of social evolution, is called eusociality.
8. the most familiar relative measure of the water content of air is relative humidity.
9. The rate at which organisms can take in energy is limited,
10. Biosphere definition
11. In the 18th century, Carolus Linnaeus published a system of taxonomy based on resemblences
12. One of the following is not how protozoa reproduce?

1 Quinine drug used for malaria
3 Horzental gene transfer.....conjunction
4 Which is not a positive interaction.....competition
5 Extension of genomics except.....vaccinomics
Food web def.

Microecology ki def

Primary succession def

Ecologist def
Resilence def
Field experiments k ponit pochy thy k kobsa nhi hai..
foundation specie ka tga
Intrinsic value ka ek tha

Ecological Genomics Definition
2. To assemble a genome from a series of sequences requires tremendous computational power.
3. The first genome to be sequenced completely was that of Haemophilus influenzae
4. The laborious edman degradation technique has now been replaced mostly by sequencing using mass spectrometry
5. Species differ from foundation species in they are more likely to belong to higher trophic levels

6. Annelids range from carnivores, herbivores, scavengers, deposit feeders, and filter feeders
8. Modern bread wheat is a mixture of recombined genes from three different wild species
9. One of the first questions that is asked when a species is considered for whole-genome sequencing is, what is the size of its genome?
10. Allelic richness Is the number of alleles in a sample (population),
11. On islands and habitat patches on continents, species richness increases with area and decreases with isolation.
14. Heterogeneity, in a community's environment, this may allow for greater species richness because there are more distinct habitats to be occupied.
16. The environment limits population growth by changing birth and death rates.
17. An environment in which resources are patchy also fosters clumped distributions.
20. The rate at which organisms can take in energy is limited,
21. Biosphere Definition
22. Multicelled algae appear as carbonaceous filaments from 1.8 billion-year-old rocks in China
23. Human is in which order: Primate
24. Maximum parsimony assumes that the tree that requires the fewest evolutionary events is most likely.
26. Oparin (1924): "Primeval soup" theory
29. Disruptive Selection: natural selection that favors two or more extreme phenotypes are favored over any intermediate phenotype
30. The advantages of in situ conservation is not one of following:
The species will have all the resources that it is adapted too
The species have more space
Bigger breeding populations can be kept
The species will not evolve in their environment
2. Vertebrates Make only ….. of all animal species? (ans) 5%
3. One of the following is not how protozoa reproduce?
4. Gymnosperm means? Ans naked seeds

1. **What are the processes of the ecosystem influenced by landscape structure? 2**

   Landscape structure influences processes such as the flow of energy, materials, and species between the ecosystems within a landscape.

2. **Define Hardy-Weinberg principle. 2**

   One of the most fundamental concepts in population genetics, the Hardy-Weinberg principle, states that "in a population mating at random in the absence of evolutionary forces, allele frequencies will remain constant."

3. **How nutrient enrichment says the human is altering the aquatic and terrestrial ecosystem? 3**

   Nutrient loss by stream ecosystems is highly pulsed and associated with disturbance by flooding. Nutrient enrichment by humans is altering aquatic and terrestrial ecosystems. ***Nitrate*** concentration and export by the earth's major rivers correlate directly with human population density. Human disturbance also increases export of ***phosphorus*** from aquatic catchments.

Nutrient enrichment appears to be reducing the diversity of plants and fungi in terrestrial ecosystems.

4. **Define and explain optical foraging theory 3**

**Optimal foraging theory attempts to model how organisms feed as an optimizing process.**

Evolutionary ecologists predict that if organisms have limited access to energy, natural selection is likely to favour individuals that are more effective at acquiring energy and nutrients. Many animals select food in a way that appears to maximize the rate at which they capture energy.

Plants appear to allocate energy to roots versus shoots in a way that increases their rate of intake of the resources that limit their growth. Plants in environments with abundant nutrients but little light tend to invest more energy in the growth of stems and leaves and less in roots. In environments rich in light but poor in nutrients, plants tend to invest more energy in the growth of roots.

5. **Lyell's Mistake on natural selection. 3**

As a part of his general resistance to direction/ substantial change in geology, Lyell maintained there was no real direction of change in the fossil record either. This was greeted with pretty general astonishment, but Lyell claimed the apparent changes were just due to our having bad (misleading) samples…more mammals would soon appear in older formations, etc., as we extend our fossil collections. Within ten years (1840) it was clear that this was completely untenable.

6. **Explain with example how exotic predators disturb the structure of food web. 5**

Exotic predators can collapse and simplify the structure of food webs. Introduced fishes have devastated the native fishes of Lake Atitlan and Gatun Lake in Central America.

**Example:**

The influence of the Nile perch on the fish community of Lake Victoria is enmeshed with massive changes in the lake's ecosystem.

Introduction of the Nile perch is rapidly reducing the species-rich fish fauna of Lake Victoria to a community dominated by a handful of species.

7. **How evolution of mutualism took place, how humans developed relationship with other species. 10**

**Evolution of Mutualism**

• Theory predicts that mutualism will evolve where the benefits of mutualism exceed the costs. **Keeler** built a cost-benefit model for the evolution and persistence of facultative plantant protection mutualisms in which the benefits of the mutualism to the plant are represented in terms of the proportion of the plant's energy budget that ants protect from damage by herbivores.

• The model assesses the costs of the mutualism to the plant in terms of the proportion of the plant's energy budget invested in extrafloral nectaries and the water, carbohydrates, and amino acids contained in the nectar. The model predicts that the mutualism will be favored where there are high densities of ants and potential herbivores and where the effectiveness of alternative defenses are low.

**Humans in Mutualism**

• Humans have developed a variety of mutualistic relationships with other species, but one of the most spectacular is that between the greater honeyguide and the traditional honey gatherers of Africa. In this apparently ancient mutualism, humans and honeyguides engage in elaborate communication and cooperation with clear benefit to both partners.

• The mutualism offers the human side a higher rate of discovery of bees' nests, while the honeyguide gains access to nests that it could not raid without human help. Careful observations have documented that the honeyguide informs the honey gatherers of the direction and distance to bees' nests as well as of their arrival at the nest.

**8. Cuvieraaa's advantage ?2**

Cuvier's anatomical and classificational work was impeccable.

He identified four basic body plans for animals– vertebrates, mollusks, arthropods, and radiate (jellyfish & others).

With these groups clearly distinguished, the chain of being was no longer part of biology.

**9. endosymbionts and evolution of eukaryotes?5**

Formed from several prokaryotic cells . In a symbiotic relationship Symbiosis involving a prolonged association of two or more dissimilar organisms, is quite common today. In many cases both symbionts benefit from the association as occurs in lichens, once thought to be plants but actually symbiotic fungi and algae

**Evidence for Endosymbiosis**

Supporting evidence for endosymbiosis comes from studies of living eukaryotic cells containing internal structures called organelles, such as mitochondria and plastids which contain their own

genetic material. In addition, prokaryotic cells synthesize proteins as a single system, whereas eukaryotic cells are a combination of protein-synthesizing systems.

**Organelles Capable of Protein Synthesis**

That is, some of the organelles within eukaryotic cells are capable of protein synthesis. These organelles with their own genetic material and protein-synthesizing capabilities are thought to have been free-living bacteria that entered into a symbiotic relationship, eventually giving rise to eukaryotic cells

## 10. Proterozoic Animal Fossils 3

- Although scarce, a few animal fossils older than those of the Ediacaran fauna are known.
- A jellyfish-like impression is present in rocks 2000 m below the Ediacara Hills Pound Quartzite.
- Burrows in many areas presumably made by worms, occur in rocks at least 700 million years old.
- Wormlike and algae fossils come from 700 to 900 million-year-old rocks in China but the identity and age of these “fossils” has been questioned